TRAUMA · NEET-SS / FET / INI-SS

Elbow and forearm

1. Distal humerus fractures (adult)

- **Two-column construct:** medial and lateral columns joined by the trochlea (the "tie-arch"). Most are **intra-articular (AO/OTA 13C)**.
- Classifications: **AO/OTA 13 A (extra-articular), B (partial articular: lateral or medial condyle, coronal shear of capitellum/trochlea), C (complete articular: C1 simple, C2 metaphyseal comminution, C3 articular comminution)**; Jupiter and Mehne (high/low T, Y, H, lambda patterns).
- **Treatment: ORIF with two plates** (orthogonal: posterolateral + medial, or **parallel: medial + lateral**, the latter biomechanically stronger), restoring articular surface, **columns and trochlear width**. O'Driscoll principles: as many long screws as possible in the distal fragments, screws passing through a plate and interdigitating, compression at the metaphysis.
- **Approach: posterior**, with **olecranon osteotomy** (**chevron, apex distal**, at the bare area (sigmoid notch), best articular exposure; risk of non-union/hardware irritation), **triceps-sparing (Alonso-Llames)**, **triceps-splitting**, or **triceps-reflecting (Bryan-Morrey, TRAP)**. **Identify and protect the ulnar nerve** (transposition is debated; many transpose anteriorly).
- **Elderly osteoporotic, comminuted, unreconstructable (C3): total elbow arthroplasty** (linked, semi-constrained; **McKee trial**: better early function than ORIF) with **lifelong lifting restrictions** (roughly 1 kg repeatedly, about 5 kg as a single lift).
- **Coronal shear fractures (capitellum/trochlea):** Bryan-Morrey (I Hahn-Steinthal: large capitellar fragment; II Kocher-Lorenz: articular shell; III comminuted; IV McKee: capitellum + trochlea), Dubberley.

Double-arc sign on lateral radiograph. Fix with **posterior-to-anterior or headless compression screws**.

- **Complications: stiffness** (commonest), **ulnar neuropathy**, heterotopic ossification, non-union (supracondylar), olecranon osteotomy non-union, infection.

2. Elbow dislocation and instability

- **Commonest adult major joint dislocation after the shoulder; posterolateral is commonest.** Mechanism: axial load, valgus, supination ("**Horii circle**": ligament failure progresses **from lateral to medial**: LUCL first, then anterior and posterior capsule, then the MCL; O'Driscoll stages).
- **Simple dislocation** (no fracture): reduce, check stability through the arc of motion, **early active motion within 1-2 weeks** (prolonged immobilisation causes stiffness; avoid over 3 weeks). Hinged brace if unstable at 30-45 degrees of extension.
- **Complex dislocation** (with fractures).
- **"Terrible triad": posterior dislocation + radial head fracture + coronoid fracture** (with LCL disruption). Treatment sequence (lateral approach, inside out): **(1) coronoid fixation (or anterior capsule repair), (2) radial head fixation or replacement (never excise), (3) LCL (LUCL) repair to the lateral epicondyle**, then **(4) MCL repair or hinged external fixator** if still unstable.
- **Varus posteromedial rotatory instability: anteromedial facet fracture of the coronoid** + LCL tear (O'Driscoll coronoid type 2); fix the anteromedial facet with a buttress plate.
- **Posterolateral rotatory instability (PLRI): LUCL insufficiency** (after dislocation, or iatrogenic after lateral surgery/tennis elbow release). Tests: **lateral pivot-shift (O'Driscoll)** (supination, valgus, axial load, flexion: reduction clunk at about 40 degrees), posterolateral drawer, chair push-up, table-top relocation. Treat with **LUCL reconstruction** (palmaris/gracilis graft).

- **Coronoid fractures: Regan-Morrey I (tip), II (up to 50%), III (over 50%)**; **O'Driscoll: tip, anteromedial facet, basal**.

3. Olecranon fractures

- Intra-articular, pulled by the triceps. **Mayo classification: I undisplaced, II displaced stable (ulnohumeral joint stable), III unstable** (A non-comminuted, B comminuted). **Schatzker** and Colton also used.
- **Treatment:** undisplaced: splint and early motion. Displaced **transverse, non-comminuted**: **tension-band wiring** (two parallel K-wires (ideally engaging the anterior ulnar cortex) + figure-of-eight wire; converts triceps tension into articular compression). **Oblique, comminuted, distal to the coronoid, or Monteggia-type**: **plate fixation** (dorsal precontoured plate). **Elderly, low-demand**: non-operative even if displaced (good function), or excision with triceps advancement (up to about 50% of the olecranon if the coronoid and anterior soft tissues are intact).
- **Complications: symptomatic hardware (commonest; removal in up to about 50-80% with TBW)**, K-wire migration, ulnar nerve symptoms, loss of extension, arthritis.

4. Radial head and neck fractures

- **Mason classification (with Johnston and Morrey modifications):**

Type	Description	Treatment
I	Undisplaced or under 2 mm	**Early mobilisation** (aspiration of haemarthrosis + local anaesthetic relieves pain and allows testing for mechanical block)
II	**Displaced over 2 mm**, partial articular (one fragment)	Non-operative if no block and stable; **ORIF (headless screws, buried in the "safe zone")** if mechanical block or displacement

Type	Description	Treatment
III	**Comminuted, whole head**	**Radial head replacement (arthroplasty)** if unreconstructable (over 3 fragments: ORIF results poor)
IV	With **elbow dislocation**	Fix or replace radial head + ligament repair

- **Safe zone for implants:** the **non-articular arc of about 90-110 degrees** that does not contact the ulna (proximal radioulnar joint): **between radial styloid and Lister's tubercle in neutral rotation** (the lateral arc in neutral).
- **Radial head is a secondary stabiliser to valgus stress** (the MCL is primary) and an important **longitudinal stabiliser** (with the interosseous membrane).
- **Never excise** the radial head when there is associated **MCL injury, coronoid fracture, interosseous membrane injury (Essex-Lopresti)** or elbow instability.
- **Radial head arthroplasty: avoid overstuffing** (gap in lateral ulnohumeral joint, capitellar wear, pain, loss of flexion): the top of the implant should be at the level of the proximal edge of the lesser sigmoid notch (about 1 mm below the coronoid tip). Metal monopolar stems (loose fit) common.
- **Essex-Lopresti injury:** radial head fracture + **interosseous membrane disruption** + **DRUJ disruption**: proximal migration of the radius (wrist pain). **Preserve or replace the radial head**; stabilise DRUJ; IOM reconstruction for chronic cases.
- **Posterior interosseous nerve** is at risk with lateral approaches: **pronate the forearm** (moves the PIN anteriorly and medially away from the field) in the **Kocher (anconeus and ECU interval)** and Kaplan (ECRB and EDC) approaches.

5. Forearm shaft fractures (adult)

- **Both-bone fractures** in adults need **ORIF with compression plates (3.5 mm LC-DCP)** restoring length, rotation and the **radial bow** (critical for forearm rotation; the **maximal radial bow** is about 60% along the radius from the proximal end).
- **Approaches: Henry (volar)** for the radius (**between brachioradialis and FCR distally; brachioradialis and pronator teres proximally**; superficial radial nerve and radial artery at risk; supinate to expose the supinator and protect the PIN proximally), **Thompson (dorsal)** between ECRB and EDC (PIN at risk proximally), ulna through the subcutaneous border (between ECU and FCU).
- **Monteggia (adult): proximal ulna fracture + radial head dislocation**. **Bado:**
  - **I** anterior dislocation, apex-anterior ulna (commonest in children).
  - **II posterior** dislocation, apex-posterior ulna (**commonest in adults**, often with radial head fracture: "**posterior Monteggia**").
  - III lateral dislocation (ulnar metaphysis).
  - IV anterior dislocation with both-bone fracture.
  - **Treatment:** anatomical **ulnar fixation (plate)**, which usually reduces the radial head; if not, open reduction (interposed annular ligament). **PIN palsy** (commonest nerve, Bado I/III) usually recovers.
- **Galeazzi: distal-third radial shaft fracture + DRUJ disruption** ("**fracture of necessity**": needs ORIF). Radial fractures **within about 7.5 cm of the articular surface** are more likely to have DRUJ instability. After plating, check the **DRUJ in supination**: stable = early motion; reducible but unstable = **pin radius to ulna in supination** or repair TFCC; irreducible = open (ECU tendon interposition).
- **Nightstick (isolated ulnar shaft) fracture:** non-operative if **under 50% displacement and under 10 degrees angulation**; otherwise plate.

- **Complications: synostosis** (single incision for both bones, high-energy, head injury, delayed surgery; excise after about 6-12 months), **refracture after plate removal** (avoid removal; protect 6 weeks), non-union, compartment syndrome, PIN injury.

> Exam pearl: **Terrible triad = posterior dislocation + radial head + coronoid; fix coronoid, radial head (fix/replace), then LCL, MCL/hinged fixator last. LUCL insufficiency = PLRI; lateral pivot-shift test.**

> Exam pearl: **Never excise the radial head with Essex-Lopresti, MCL or coronoid injury. Avoid overstuffing radial head arthroplasty. Pronate the forearm to protect the PIN in lateral approaches. Galeazzi = fracture of necessity; Bado II posterior Monteggia is commonest in adults.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.